Mixtures of Deterministic Programs in Probabilistic Coherence Spaces

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Abstract. Probabilistic coherence spaces model higher-order probabilistic programs. We ask which elements of these spaces are mixtures of deterministic programs, the results of choosing a deterministic program at random. For each type of multiplicative–additive linear logic, we call the convex hull of its deterministic elements its deterministic hull. Every coherence space lies between its deterministic hull and a local relaxation cut out by the deterministic elements of the dual type. The orthogonal of the deterministic hull of a type is the local relaxation of its dual. We locate the coherence space between these two bounds. At first order all three sets coincide. At the type of a program that calls two functions, the coherence space is the whole local relaxation. When the data types have at least two elements, it contains behaviours that no randomized sequential program has, so some laws of sequential programs fail in the model. At the dual type the coherence space is the deterministic hull. Its gap to the local relaxation is Bell's theorem. For duals of tensors of additive types, induced four-cycles in the coherence graphs decide whether a gap occurs. The gap then transfers to every formula over nonempty data types in which such a tensor occurs negatively. The main results are mechanized in Lean 4.

Keywords: Probabilistic coherence spaces · Linear logic · Probabilistic programs · Perfect graphs · Contextuality · Lean

1 Introduction

A randomized program can be run by choosing a deterministic program at random and running it. For first-order while programs, this gives the same semantics as linear operators on measures [37]. We ask whether it holds at higher types.

Probabilistic coherence spaces are a model of linear logic in which a type denotes a set of weight vectors on a set of points, its web [25,18]. They are fully abstract for probabilistic PCF, the extension of PCF (Programming Computable Functions) with probabilistic choice [20]. The proof of full abstraction uses the form of the morphisms as power series, and not the definability of every element of a space. In a concurrent game model of probabilistic PCF, definability holds for the strategies that call one argument at a time and depend only on the history they observe [13]. We ask which elements of a space are mixtures of its elements with values in $\{0, 1\}$. Full abstraction does not settle this question, since

Table 1. The coherence space $P(A)$ between the deterministic hull and the local relaxation of A. Strictness needs data types with at least two elements, and in the last row it is shown over Bool.

A	hull and $P(A)$	$P(A)$ and relaxation	Sections
$\mathbf{n} \multimap \mathbf{m}$ (first order)	equal	equal	3, 7.3
T (second order)	strictly smaller	equal	4–6, 7.1
dual of T	equal	strictly smaller	5.2, 7.4
$(T \otimes T)^{\perp}$	contained	strictly smaller	7.2

it concerns the equivalence of programs and not the laws valid on a whole space. Such laws bound the behaviour of every program of a type at once. They can fail on a space that contains more than the mixtures of deterministic programs.

We call these elements the *certain states* of a type. They are its deterministic elements, and the set of their mixtures is their convex hull, so we call it the *deterministic hull* of the type. This is the construction of generalized probabilism [43,53], applied to the points of a web. We work in multiplicative–additive linear logic (MALL) over finite data types, whose programs use each argument exactly once. At every MALL formula the deterministic hull lies inside the coherence space. Each certain state is a clique of Girard's coherence space. A *gap* is an element of the coherence space outside the deterministic hull. It violates a linear inequality that every certain state satisfies [21,43].

The coherence space also has an upper bound defined from certain states. The *local relaxation* of a formula is the set of weight vectors with total weight at most 1 on the support of every certain state of its negation. It contains the coherence space. Its orthogonal is the deterministic hull of the negation. So every coherence space lies between two sets defined from certain states, and Table 1 shows where.

At first order the three sets coincide: a probabilistic program from one data type to another is a mixture of deterministic programs. At the type T of a program that calls two functions once each, the deterministic hull is the set of randomized sequential programs, by Kuhn's theorem [38]. It is the stable set polytope of a graph whose clique-constrained polytope is the coherence space. The coherence space is the whole local relaxation. When the data types have at least two elements, the function with value $\frac{1}{2}$ on five points of the web lies in the coherence space and violates the Klyachko–Can–Binicioğlu–Shumovsky (KCBS) inequality [33], which every randomized sequential program satisfies. So every law valid on the coherence space of T holds for sequential programs, but not conversely.

At the dual of T, the tensor of the two argument types, the picture is reversed. The coherence space is the deterministic hull. When the data types have at least two elements, it lies strictly inside the local relaxation. The Popescu–Rohrlich box [47] lies in the local relaxation and outside the coherence space, so this gap is Bell's theorem. The two gaps are equivalent by the duality above. The deterministic hulls of a formula and of its negation form a tight pair, each the

orthogonal of the other, iff neither has a gap. Tightness holds at first order and can fail at a tensor of first-order types. So the deterministic hull is not compositional.

For the dual of a tensor of two formulas built from data types with & and $\oplus$, the gap is governed by induced four-cycles of the coherence graphs. The direction without a gap has an explicit decomposition, mechanized for formulas built from the unit type without perfect graph theory. Coordinate retracts, embeddings of webs that preserve the coherence spaces in both directions, carry the gap to every formula over nonempty data types in which such a tensor occurs negatively. In quantum event logic no state has values in $\{0, 1\}$ [36], so only the local relaxation applies.

Contributions.

1. **Cliques.** Every certain state of a MALL formula is a clique of its coherence space, but not conversely (Theorems 1 and 10).
2. **Second order.** At T the deterministic hull and the coherence space are the stable set and the clique-constrained polytopes of one graph, with explicit gaps for two functional arguments and, among the always-terminating programs, for three (Theorems 4 and 3, Proposition 2).
3. **Exact condition.** For formulas built from data types with & and $\oplus$, the dual of a tensor has no gap iff one factor has no induced four-cycle, with a direct proof mechanized over the unit type (Theorem 5).
4. **Transfer.** Coordinate retracts carry the gap to every formula over nonempty data types with such a tensor occurring negatively, a condition not necessary beyond second order (Theorem 8, Proposition 3).
5. **Duality.** The deterministic hulls of a formula and of its negation are tight iff neither has a gap, which holds at first order and can fail at a tensor of first-order types (Theorem 11, Corollary 3).
6. **Bell.** At the dual of T, the Popescu–Rohrlich box lies in the local relaxation and outside the coherence space (Theorem 12).

Every result of the paper is mechanized in Lean 4 over mathlib [42,51], in about 8,300 lines, except steps the text marks as not formalized. A superscript number such as $^{\text{L1}}$ links a statement to its numbered Lean counterpart. The supplementary material lists the Lean statements and describes the development.

2 Mixtures of Certain States

This section describes a construction that takes a set of events and a set of assignments that settle every event, and returns the set of mixtures of these assignments [21,43,53]. Later sections apply it to the types of linear logic. All functions take values in the nonnegative reals $\mathbb{R}_{\geq 0}$.[1]

[1] The Lean development works in the extended nonnegative reals $[0, \infty]$ of mathlib, the type in which measures take values. Every function in this paper takes finite values, so the statements are the same, except that some Lean statements carry a finiteness hypothesis that is automatic over $\mathbb{R}_{\geq 0}$.

Definition 1 (presentation, deterministic hull[L1]).

(i) For a given set E, whose elements are the events*, the* certain states *of a* presentation *(E, S) are the elements of a set S of functions $E \to \mathbb{R}_{\geq 0}$ with values in $\{0, 1\}$. For $x : E \to \mathbb{R}_{\geq 0}$ and $e \in E$ we write x_e for $x(e)$, the weight that x gives to e.*

(ii) For given functions $s_1, \ldots, s_n : E \to \mathbb{R}_{\geq 0}$, the mixtures *of them are the functions $\sum_i w_i\, s_i$ such that $w_i \in \mathbb{R}_{\geq 0}$ and $\sum_i w_i = 1$.*

(iii) For a given presentation (E, S), the deterministic hull $\mathrm{Mix}(S)$ *is the set of mixtures of certain states.*

(iv) For a given set C of functions $E \to \mathbb{R}_{\geq 0}$, C is closed under mixtures *if it contains every mixture of its elements.*

A certain state settles every event, with value 1 or 0, so the certain states are the deterministic assignments and $\mathrm{Mix}(S)$ is their convex hull, hence its name. A mixture $\sum_i w_i s_i$ is the result of choosing a certain state at random, s_i with probability w_i: its value at an event e is the probability that the chosen state gives e the value 1. The deterministic hull has an axiomatic description by linear inequalities.

Definition 2 (affine law[L2]). *For a given finite set E, the* affine laws *on E are the inequalities $\sum_{e \in E} a_e x_e + k \leq \sum_{e \in E} b_e x_e + k'$ such that $a, b : E \to \mathbb{R}_{\geq 0}$ and $k, k' \in \mathbb{R}_{\geq 0}$. The functions that* satisfy *a law are the $x : E \to \mathbb{R}_{\geq 0}$ such that the inequality is true at x, and a law is* valid on *a set of functions if every function in the set satisfies it.*

For finite E, $\mathrm{Mix}(S)$ is the set of functions that satisfy every affine law valid on S, by the completeness theorem of generalized probabilism [21,43][L3,L4]. We call the affine laws valid on S the *laws of the presentation*. The presentation is *sound* for a set C of functions if every law of the presentation is valid on C [39].

Definition 3 (gap). *For a given presentation (E, S), the sets with a* gap *over S are the sets C of functions $E \to \mathbb{R}_{\geq 0}$ such that C contains S, is closed under mixtures, and is not* $\mathrm{Mix}(S)$.

A set that contains S and is closed under mixtures contains $\mathrm{Mix}(S)$, so a gap is a strict inclusion. An element of a gap does not arise from choosing a certain state at random. The completeness theorem allows us to characterize gaps by laws.

Corollary 1 (a gap is a violated law[L4]). *Let E be finite, and let C be a set of functions $E \to \mathbb{R}_{\geq 0}$ that contains S and is closed under mixtures. Then C has a gap over S iff some element of C violates a law of the presentation, that is, iff the presentation is not sound for C.*

A simple form of law bounds how many events of a finite set $F \subseteq E$ a certain state can give the value 1. If no certain state gives the value 1 to more than k events of F, then $\sum_{e \in F} x_e \leq k$ is a law of the presentation, a *counting law*. Many Bell and noncontextuality inequalities, such as the Clauser–Horne–Shimony–Holt (CHSH) and the KCBS inequalities, have this form [44,12], and every gap found in this paper violates a counting law.

3 Linear Logic

In probabilistic coherence spaces a formula denotes a set of weight vectors on a set of points [18]. The construction of Section 2 applies to them, with the points as events and the elements with values in $\{0,1\}$ as certain states.

3.1 Coherence Spaces, Data Types, First Order

Definition 4 (probabilistic coherence space[L6]).

(i) For a given finite set W, the web*, whose elements are the* points*, the* weight vectors *on W are the functions $W \to \mathbb{R}_{\geq 0}$, and their pairing is $\langle x, y\rangle = \sum_{w\in W} x_w y_w$.*

(ii) For a given set Q of weight vectors, the orthogonal *$Q^\perp$ is the set of weight vectors y such that $\langle x, y\rangle \leq 1$ for every $x \in Q$, and $Q^{\perp\perp}$ is the* biorthogonal *of Q.*

(iii) For a given web W, the probabilistic coherence spaces *(PCS) with web W are the sets P of weight vectors on W such that $P = P^{\perp\perp}$, every point w has some $x \in P$ with $x_w \neq 0$, and for every point w the values x_w with $x \in P$ are bounded [18].*

Every orthogonal is closed under mixtures, since $\langle x, \sum_i w_i y_i\rangle = \sum_i w_i \langle x, y_i\rangle \leq \sum_i w_i = 1$[L14]. A PCS is the orthogonal of $P^\perp$, so it is closed under mixtures.

For a finite set X, the *data type* X is the PCS of the $x : X \to \mathbb{R}_{\geq 0}$ with $\sum_{a\in X} x_a \leq 1$, the subprobability distributions on X[L6]. An element is the output distribution of a computation with values in X, and $1 - \sum_a x_a$ is the probability that it does not terminate. We write $\mathbf{n}$ for the data type $\{1, \ldots, n\}$, and Bool for $\mathbf{2}$ with values t and f. For data types X and Y, the PCS $X \multimap Y$ on the web $X \times Y$ is the set of $t : X \times Y \to \mathbb{R}_{\geq 0}$ with $\sum_b \sum_a x_a\, t(a, b) \leq 1$ for every x in X, equivalently $\sum_b t(a, b) \leq 1$ for every $a \in X$[L5]. These are the substochastic matrices, the probabilistic programs from X to Y that may not terminate, with $t(a, b)$ the probability of output b on input a.

A *coherence space* is a set with a reflexive and symmetric relation, its *coherence* [24]. A *clique* is a set of pairwise coherent elements. In the coherence space $X \multimap Y$, two points (a, b) and (a', b') cohere iff they are equal or $a \neq a'$. So the cliques are the graphs of partial functions from X to Y.

The implications $X \multimap Y$ between data types are the *first-order* types. Take as certain states the elements of $X \multimap Y$ with values in $\{0, 1\}$. Write $Y_\perp$ for Y with an added value $\perp$ for non-termination. For $f : X \to Y_\perp$, let $\det f(a, b) = 1$ if $f(a) = b$ and 0 otherwise. We show that the certain states are these deterministic programs and that $X \multimap Y$ has no gap over them.

Proposition 1 (first order[L7,L8]). *Let X and Y be finite. (i) $X \multimap Y = \mathrm{Mix}\{\det f \mid f : X \to Y_\perp\}$. (ii) A function $t : X \times Y \to \mathbb{R}_{\geq 0}$ with values in $\{0, 1\}$ lies in $X \multimap Y$ iff its support is a clique of the coherence space $X \multimap Y$.*

Both parts are elementary. For (i), each row $t(a, \cdot)$ is a subprobability distribution on Y, and so a mixture of the point masses and the zero vector. Choosing one of these independently for each row, with the product of the weights, allows for expressing t as a mixture of deterministic programs. For (ii), a substochastic matrix with values in $\{0, 1\}$ has at most one entry 1 in each row.

3.2 MALL Formulas and Their Semantics

Multiplicative–additive linear logic (MALL) builds formulas from data types with linear negation and two conjunctions [24]. The formula $A \otimes B$ is the type of a pair of independent elements of A and of B. The formula $A \,\&\, B$ is the type of an element that can be used as an element of A or of B, at the choice of its user.

Definition 5 (formulas and their semantics[L9]).

(i) The formulas *are the terms generated from the data types* $\mathbf{n}$*, for* $n \in \mathbb{N}$*, by linear negation* $A^\perp$*, the tensor* $A \otimes B$*, and the product* $A \,\&\, B$*. The derived connectives are* $A \parr B = (A^\perp \otimes B^\perp)^\perp$*,* $A \oplus B = (A^\perp \,\&\, B^\perp)^\perp$*, and* $A \multimap B = (A \otimes B^\perp)^\perp$*.*

(ii) For a given formula A*, the* web $|A|$ *is* $\{1, \ldots, n\}$ *for* $\mathbf{n}$*,* $|A|$ *for* $A^\perp$*,* $|A| \times |B|$ *for* $A \otimes B$*, and the disjoint union* $|A| \sqcup |B|$ *for* $A \,\&\, B$*.*

(iii) For a given formula A*, the PCS* $P(A)$ *is the data type* $\mathbf{n}$ *for* $\mathbf{n}$*,* $P(A)^\perp$ *for* $A^\perp$*,* $\{x \otimes y \mid x \in P(A),\ y \in P(B)\}^{\perp\perp}$ *for* $A \otimes B$*, where* $(x \otimes y)(a, b) = x_a y_b$*, and the set of* z *whose restrictions to* $|A|$ *and* $|B|$ *lie in* $P(A)$ *and* $P(B)$ *for* $A \,\&\, B$ *[18].*

(iv) For a given formula A*, the* coherence $\asymp_A$ *is the relation on* $|A|$ *such that* $s \asymp_{\mathbf{n}} t$ *iff* $s = t$*,* $s \asymp_{A^\perp} t$ *iff* $s = t$ *or not* $s \asymp_A t$*, two pairs cohere in* $A \otimes B$ *iff their first components cohere and their second components cohere, and points of* $A \,\&\, B$ *cohere iff they cohere in* A *or in* B*, or lie in different parts [24]. The* coherence graph G_A *is the graph on* $|A|$ *whose edges are the pairs of distinct coherent points.*

The semantics computes the spaces of Section 3.1: $P(\mathbf{n} \multimap \mathbf{m})$ is the PCS $\{1, \ldots, n\} \multimap \{1, \ldots, m\}$[L15], with the coherence described there. So Proposition 1 is a statement about the formulas $\mathbf{n} \multimap \mathbf{m}$[L16].

Each formula A gives a presentation. Its events are the points of $|A|$, and its certain states are the elements of $P(A)$ with values in $\{0, 1\}$. The *deterministic hull* of A is Mix of these certain states. Every $P(A)$ is a PCS [18][L10], so it contains the deterministic hull of A[L13]. Definition 3 therefore applies to it. We say that A has a *gap* if $P(A)$ has a gap over the certain states of A, that is, if the presentation of A is not sound for $P(A)$ (Corollary 1).

3.3 Coherence at Every Formula

We prove that the coherence space bounds the weights of pairs of points, and so constrains the certain states.

Theorem 1 (coherence at every formula[L11,L12]). *Let A be a formula. The indicator of every point and of every pair of distinct coherent points lies in $P(A)$. Every $x \in P(A)$ satisfies $x_s \leq 1$ for every point s, and $x_s + x_t \leq 1$ for every pair of distinct incoherent points s, t. In particular, every certain state of A is the indicator of a clique of the coherence space A.*

The proof is a simultaneous induction on A. Negation exchanges the statement about coherent pairs with the statement about incoherent pairs. For $A \otimes B$, the indicator of a coherent pair is bounded by the tensor of the indicators of its two projections. A pair incoherent in $A \otimes B$ is incoherent in one component, where the bound of A or of B applies together with the bound on single points. So the bound on incoherent pairs, the motivating intuition for probabilistic coherence spaces [25], holds at every multiplicative–additive formula over data types.

4 Second Order

At first order there is no gap (Proposition 1). This section shows that there is one at the second-order type of a program with two functional arguments, and describes the deterministic hull there.

4.1 Sequential Programs and Kuhn's Theorem

Let X, Y, X', Y' be data types. The formula $K = (X \multimap Y) \otimes (X' \multimap Y')$ is the *core*. Its negation $T = K^{\perp}$ is the type of a program that receives two programs $f : X \multimap Y$ and $g : X' \multimap Y'$, calls each once, and then terminates or does not. Its web is $(X \times Y) \times (X' \times Y')$. A point $((a, b), (c, d))$ records that the program called f on a and received b, and called g on c and received d. By Definition 5(iii), $P(T)$ is the orthogonal of the products: it is the set of t with $\sum_{p,q} y_p \, t(p, q) \, z_q \leq 1$ for all $y \in P(X \multimap Y)$ and $z \in P(X' \multimap Y')$[L29]. For arguments y and z, the sum $\sum_{p,q} y_p \, t(p, q) \, z_q$ is the probability that t terminates.

A *randomized sequential program* of type T calls one argument first. If it calls f first, it chooses the input a with probability $\mu(a)$, after the answer b chooses the input c to g with probability $\kappa(a, b, c)$, and after the answer d terminates with probability $\rho(a, b, c, d)$. Its value at $((a, b), (c, d))$ is $\mu(a) \, \kappa(a, b, c) \, \rho(a, b, c, d)$ plus the analogous term for the branch that calls g first, with $\sum \mu \leq 1$ over both branches, $\sum_c \kappa(a, b, c) \leq 1$, and $\rho \leq 1$. It is *deterministic* if its values are in $\{0, 1\}$. We prove the following, whose part (ii) is an instance of Kuhn's theorem [38].

Theorem 2 (second order[L17,L18,L19]). *Let X and X' be nonempty. (i) A function with values in $\{0, 1\}$ lies in $P(T)$ iff its support lies in a set $\{((a, b), (c(b), d)) \mid b \in Y,\ d \in Y'\}$ for some $a \in X$ and $c : Y \to X'$, or in a set $\{((a(d), b), (c, d)) \mid b \in Y,\ d \in Y'\}$ for some $c \in X'$ and $a : Y' \to X$. These are exactly the deterministic sequential programs. (ii) The deterministic hull of T is the set of randomized sequential programs.*

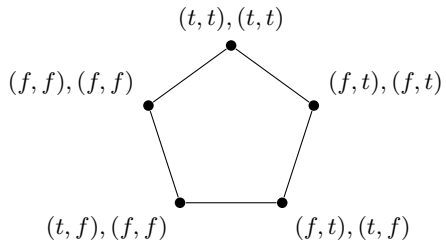


Fig. 1. The five points of Theorem 3 over Bool, where $(a,b),(c,d)$ records $f(a)=b$ and $g(c)=d$. Two points are joined when one pair of deterministic programs produces both. The graph has no triangle, so the function with value $\frac{1}{2}$ on each point lies in $P(T)$. It has no three pairwise non-adjacent points, so every randomized sequential program has total weight at most 2 on them.

The two sets in (i) are the decision trees of a program that calls f first and of one that calls g first. Part (ii) has two directions. Every randomized sequential program is a mixture of deterministic ones[L20]. Every mixture of randomized sequential programs is a randomized sequential program[L21].

For nonempty X and X', we also prove that the deterministic hull of T is the set of denotations of a small language[L22,L25]. A program first calls f or g on a fixed input, continues for each answer with a call of the other function, and then terminates or not, with a finite probabilistic choice $\mathsf{choose}(w_1P_1,\ldots,w_kP_k)$, $\sum_i w_i \leq 1$, at every stage. Every program denotes a randomized sequential program[L23]. Every certain state of T is a denotation[L24]. The language has no recursion. We do not relate it to probabilistic PCF at this type, or the randomized sequential programs to the sequential strategies of [13].

4.2 The Pentagon Gap

The deterministic programs of type $X \multimap Y$ are its certain states, the indicators of the graphs of partial functions. A pair (f,g) of deterministic programs *produces* the point $((a,b),(c,d))$ if $f(a)=b$ and $g(c)=d$. Two distinct points of the web of T are *exclusive* if one pair of deterministic programs produces both, that is, if they cohere in the core K. No certain state of T gives the value 1 to two exclusive points, since its pairing with the product of that pair of programs would be at least 2. The *exclusivity graph* of T has the points of the web as vertices and the exclusive pairs as edges. We show that T has a gap.

Theorem 3 (the pentagon[L26,L27,L28]). *Let X,Y,X',Y' have at least two elements each. There are five points of the web of T such that the function with value $\frac{1}{2}$ on them and 0 elsewhere lies in $P(T)$ and not in the deterministic hull of T. For $X=Y=X'=Y'=\mathrm{Bool}$, every randomized sequential program has total weight at most 2 on the five points, every element of $P(T)$ has total weight at most $\frac{5}{2}$ on them, and both bounds are attained.*

Over Bool, write (a, b) for the observation $f(a) = b$. The five points are $((t,t),(t,t))$, $((t,f),(f,f))$, $((f,t),(t,f))$, $((f,t),(f,t))$, and $((f,f),(f,f))$. They induce a cycle of length 5 in the exclusivity graph (Fig. 1). The proof uses that this induced subgraph has no triangle and no three pairwise non-adjacent vertices, which Lean checks by evaluation. It embeds the five points into larger types by injections from Bool. So no certain state gives the value 1 to three of the five points. Hence T has a gap: the function of the theorem violates the counting law that bounds the total weight on the five points by 2.

The bounds 2 and $\frac{5}{2}$ are the independence number of the cycle C_5 of length 5, the largest number of pairwise non-adjacent vertices, and its fractional relaxation. They are the noncontextual and the exclusivity bounds of the KCBS scenario [12,33].

4.3 Divergence and Total Programs

The function of Theorem 3 is partial: for some pairs of total functions it terminates with probability less than 1, for instance with probability $\frac{1}{2}$ for f the negation and g the identity on Bool. At T a gap needs such partiality. Let $t \in P(T)$ terminate with probability 1 on every pair of total functions. Constant arguments show that t is a conditional distribution of the inputs (a, c) given the answers (b, d). Termination with probability 1 for all total functions makes it a bipartite classical process without predefined causal order [5]. Every such process is a mixture of processes in which one party acts first [5], and these are randomized sequential programs. So t lies in the deterministic hull of T by Theorem 2(ii). This argument is not formalized. With three functional arguments, we find a gap among the elements that terminate with probability 1 on every triple of total functions.

Proposition 2 (a gap among total programs[L30]). *Let* $T_3 = ((\mathbf{2} \multimap \mathbf{2}) \otimes (\mathbf{2} \multimap \mathbf{2}) \otimes (\mathbf{2} \multimap \mathbf{2}))^{\perp}$. *Some element of* $P(T_3)$ *terminates with probability* 1 *on every triple of total functions and is not in the deterministic hull of* T_3.

The element has value 1 at one point of the web and $\frac{1}{2}$ at fourteen others. Since neither $\mathbf{2} \multimap \mathbf{2}$ nor its tensor square has a gap, membership reduces to the 64 triples of total functions (proof of Theorem 5). On each of them the element has total weight exactly 1. Five of its points with value $\frac{1}{2}$ form a cycle of length 5 of exclusive pairs. So, as for Theorem 3, every mixture of certain states has total weight at most 2 on them, against $\frac{5}{2}$. By the argument above, the element is a tripartite classical process [5] that is not a mixture of deterministic ones. This identification is not formalized.

5 The Exact Condition

This section describes the deterministic hull and the coherence space of T as two polytopes of one graph, and gives an exact condition for a gap at the dual of a tensor of two formulas.

5.1 STAB and QSTAB

For a finite graph G with vertex set V, a *clique* is a set of pairwise adjacent vertices and a *stable set* a set of pairwise non-adjacent vertices. The *stable set polytope* $\mathrm{STAB}(G)$ is Mix of the indicators of the stable sets, and the *clique-constrained polytope* $\mathrm{QSTAB}(G)$ is the set of $x : V \to \mathbb{R}_{\geq 0}$ with $\sum_{v \in C} x_v \leq 1$ for every clique C [28]. A stable set meets every clique in at most one vertex, so $\mathrm{STAB}(G) \subseteq \mathrm{QSTAB}(G)$. We prove that the two polytopes describe the coherence space and the deterministic hull of T.

Theorem 4 (graph polytopes[L31,L32]). *Let G be the exclusivity graph of T. Then $P(T) = \mathrm{QSTAB}(G)$, and the deterministic hull of T is $\mathrm{STAB}(G)$.*

Every element of $P(X \multimap Y)$ is a mixture of certain states (Proposition 1). Every certain state lies below the indicator of the graph of a total function. So $P(T)$ is cut out by the products of two such indicators. Their supports are the maximal cliques of G, so $P(T) = \mathrm{QSTAB}(G)$. An element of $\mathrm{QSTAB}(G)$ with values in $\{0, 1\}$ is the indicator of a stable set, so the deterministic hull is $\mathrm{STAB}(G)$.

So T has a gap iff $\mathrm{STAB}(G) \neq \mathrm{QSTAB}(G)$. A graph is *perfect* if every induced subgraph has chromatic number equal to its largest clique size. Then $\mathrm{STAB}(G) = \mathrm{QSTAB}(G)$ iff G is perfect [40,14]. The cycle of length 5 of Theorem 3 is an induced subgraph of G that is not perfect.

5.2 The Clique Property and Perfection

A formula X has the *clique property* if $P(X)$ is the set of mixtures of the indicators of the cliques of its coherence graph G_X. Data types, their negations, and the formulas $\mathbf{n} \multimap \mathbf{m}$ (Proposition 1) have it. So does every formula built from data types with & and $\oplus$, since $P(A \,\&\, B)$ is the product of $P(A)$ and $P(B)$ and $P(A \oplus B)$ is the convex hull of the copies of $P(A)$ and $P(B)$ on the two parts of the web [18]. Lean checks this for the formulas built from the data type $\mathbf{1}$ with & and $\oplus$[L35]. Their coherence graphs are the *cographs*, the graphs with no induced path on four vertices [16], with & as join, which adds all edges between the two parts, and $\oplus$ as disjoint union.

An *induced 4-cycle* of a formula X is a quadruple of points a, b, c, d with $a \asymp b \asymp c \asymp d \asymp a$ and with a, c and b, d incoherent. The *strong product* $G \boxtimes H$ of two graphs has the pairs of vertices as vertices, and two distinct pairs are adjacent iff each coordinate is equal or adjacent. By Definition 5(iv), the coherence graph of $X \otimes Y$ is $G_X \boxtimes G_Y$. We prove the following exact condition.

Theorem 5 (second order[L33,L34,L40]). *Let X and Y have the clique property. (i) $X \otimes Y$ has no gap. Moreover, $(X \otimes Y)^\perp$ has no gap iff $G_X \boxtimes G_Y$ is perfect. (ii) If G_X and G_Y are cographs, then $(X \otimes Y)^\perp$ has no gap iff G_X or G_Y has no induced 4-cycle.*

If neither X nor Y has a gap, then neither do $X \otimes Y$, $X \,\&\, Y$, and $X \oplus Y$. Moreover, $P((X \otimes Y)^\perp)$ is cut out by the products of the certain states of X

and of Y. The Lean proof shows a bipolar theorem for downward-closed sets of $\{0,1\}$-valued functions by hyperplane separation, as for the completeness theorem of Section 2. With the clique property these products are the indicators of the products $K \times L$ of cliques, which are the maximal cliques of $F = G_X \boxtimes G_Y$. So $P((X \otimes Y)^\perp) = \mathrm{QSTAB}(F)$, and its certain states are the indicators of the stable sets of F. The deterministic hull is therefore $\mathrm{STAB}(F)$, which equals $\mathrm{QSTAB}(F)$ iff F is perfect. In (ii), the gap when both graphs have an induced 4-cycle follows from Theorem 8, which holds for all formulas. The closure statements, the description of $P((X \otimes Y)^\perp)$ as the orthogonal of the products of certain states, and (ii) for X and Y built from $\mathbf{1}$ with & and $\oplus$ are formalized. The description as $\mathrm{QSTAB}(F)$ is formalized at T (Theorem 4), and the equivalence with perfection in (i) is cited.

For $X = \mathbf{n} \multimap \mathbf{m}$ the coherence graph is complete multipartite, with the inputs as the n classes of m points each and two points adjacent iff they lie in different classes. It has an induced 4-cycle iff $n, m \geq 2$: two points in each of two classes form one, for $n = 1$ the graph has no edges, and for $m = 1$ it is complete. So T has a gap iff each of X, Y, X', Y' has at least two elements, and the core has no gap[L34]. The gap when all four have at least two elements follows from Theorem 3. The absence of a gap otherwise uses (ii) for formulas not built from $\mathbf{1}$, through the transfer below, which is not formalized.

5.3 Four-Cycles: a Direct Proof

We prove the direction of (ii) without a gap by an explicit decomposition, without perfect graph theory. If neither side of $A \,\&\, B$ is a clique, an incoherent pair in A and one in B form an induced 4-cycle. So a formula built from $\mathbf{1}$ with & and $\oplus$ that has no induced 4-cycle is isomorphic, by a bijection of the webs that preserves coherence in both directions, to one built from $\mathbf{1}$ with $\oplus$ and $A \mapsto A \,\&\, \mathbf{1}$[L36]. Its points are the nodes of a rooted forest, and two points cohere iff one lies on the path from a root to the other. So G_X is the comparability graph of the forest [54,27]. For $v : |Y| \to \mathbb{R}_{\geq 0}$ let $\omega(v)$ be the largest total weight of a clique of G_Y. Then z lies in $P((X \otimes Y)^\perp)$ iff $\omega(\sum_{x \in \pi} z(x, \cdot)) \leq 1$ for every path π from a root[L37]. Every such z is a mixture of $\{0,1\}$-valued functions whose sums along the paths are indicators of stable sets of G_Y[L38]. The proof is an induction on the forest, whose step at a root is an induction on Y. At $\oplus$ the two components are decomposed separately. At & a stable set lies in one component, and is drawn there with probability given by ω, after a split of the weights below the root that keeps ω within its level along every path[L39].

Lean checks (ii) in both directions for X and Y built from $\mathbf{1}$ with & and $\oplus$[L40]. For the other formulas in (ii) with nonempty webs, the clique property makes a coherence-preserving bijection onto such a formula preserve the coherence spaces. This transfer is not formalized. With the cited part of (i), the proof shows that the strong product of a forest comparability graph and a cograph is perfect. This consequence is not formalized. This extends the theorem that the strong product of two forest comparability graphs, the graphs with no induced path on four

vertices and no induced 4-cycle, is perfect [48]. We did not find the extension in the literature.

5.4 Beyond Second Order

An occurrence of a formula D in a formula A is *positive* if it lies under an even number of negations and *negative* otherwise, the derived connectives $\parr$, $\oplus$, and $\multimap$ being unfolded. So the left argument of $\multimap$ is a negative position. Beyond second order, we find a gap even when no negatively occurring tensor has an induced 4-cycle in both factors.

Proposition 3 (a gap without four-cycles[L41]). *The formula* $(\mathbf{2} \,\&\, \mathbf{1}) \multimap ((\mathbf{2} \multimap \mathbf{2}) \otimes (\mathbf{2} \multimap \mathbf{2}))$ *has a gap. Every tensor occurring negatively in it has a factor with at most three points, so none has an induced* 4*-cycle in both factors.*

The formula is $((\mathbf{2} \,\&\, \mathbf{1}) \otimes T)^{\perp}$. An element outside the deterministic hull has value $\frac{1}{2}$ at five points of the web. Its membership in the coherence space uses the bounds of Theorem 1 at three incoherent pairs and at one point of T. The five points form an induced cycle of length 5 in the coherence graph, so by Theorem 1 every certain state is 1 at no more than two of them. So every mixture has total weight at most 2 there, against $\frac{5}{2}$. The second statement of the proposition is checked by inspection of the formula.

6 Other Formulas

A gap transfers along a kind of embedding of webs. This section uses it to carry the gap of Theorem 3 to every formula with a negative occurrence of a suitable tensor.

6.1 Coordinate Retracts

Definition 6 (coordinate retract[L42]). *For given formulas D and A, the* coordinate retracts *of D into A are the injective maps $j : |D| \to |A|$ such that the pushforward j_*, defined by $(j_*x)(j(d)) = x_d$ and value 0 outside the image of j, maps $P(D)$ into $P(A)$, and the pullback $z \mapsto z \circ j$ maps $P(A)$ into $P(D)$. We say that D is a coordinate retract of A if such a map exists.*

The pullback of j_*x is x, and this allows us to prove that gaps transfer along coordinate retracts.

Theorem 6 (transfer[L43,L44]). *(i) If D is a coordinate retract of A and D has a gap, then A has a gap. (ii) Coordinate retracts compose. If D is a coordinate retract of A, then $D^{\perp}$ is a coordinate retract of $A^{\perp}$. The formula D is a coordinate retract of $D \,\&\, B$, of $B \,\&\, D$, and, if the web of B is nonempty, of $D \otimes B$ and of $B \otimes D$.*

For (i), take $x \in P(D)$ outside the deterministic hull of D. If j_*x were a mixture of certain states of A, pulling the mixture back along j would write $x = (j_*x) \circ j$ as a mixture of certain states of D. For negation, the pairing satisfies $\langle j_*u, z\rangle = \langle u, z \circ j\rangle$, which exchanges the two conditions of Definition 6. For $D \otimes B$, the map is $j(d) = (d, b)$ for a point b of B, so $j_*x = x \otimes \delta_b$, where δ_b is the indicator of b. The pullback condition uses that every element of $P(B)$ is at most 1 at b (Theorem 1).

6.2 Negative Occurrences and Four-Cycles

With positive and negative occurrences as defined in Section 5.4, we prove the following.

Theorem 7 (the gap at other formulas[L45]). *Let $n, m, n', m' \geq 2$, and let A be a formula all of whose data types are nonempty. If $(\mathbf{n} \multimap \mathbf{m}) \otimes (\mathbf{n}' \multimap \mathbf{m}')$ occurs negatively in A, then A has a gap.*

A positive occurrence of D gives a coordinate retract of D into A, and a negative one a coordinate retract of $D^\perp$, by Theorem 6(ii) and induction on A, using that D is a coordinate retract of $D^{\perp\perp}$. Theorem 6(i) and Theorem 3 then give the claim. Examples are T itself, the type $((\mathbf{n} \multimap \mathbf{m}) \otimes (\mathbf{n}' \multimap \mathbf{m}')) \multimap B$ of a program that receives two functions and returns a value of type B, for every B with nonempty data types (Lean checks the instance $B = \mathbf{2}$). A positive occurrence of the core does not suffice: the core itself has no gap (Theorem 5(i)).

The occurrence condition of Theorem 7 refers to one syntactic subformula. We also prove a version that depends only on the coherence graphs of the factors.

Theorem 8 (four-cycles[L46,L47,L48]). *If X and Y both have an induced 4-cycle, then $(X \otimes Y)^\perp$ has a gap, and so does every formula over nonempty data types in which $X \otimes Y$ occurs negatively.*

The four points of an induced 4-cycle of X form a coordinate retract of $\mathbf{2} \multimap \mathbf{2}$ into X, with input 0 sent to a and c and input 1 to b and d. The pushforward lands in $P(X)$ because the indicators of the coherent pairs of the cycle lie in $P(X)$ (Theorem 1). The pullback lands in $P(\mathbf{2} \multimap \mathbf{2})$ because the incoherent pairs have total weight at most 1 (Theorem 1). Coordinate retracts pass through $\otimes$, so Theorems 3 and 6 give the claim. The coherence graph of $\mathbf{n} \multimap \mathbf{m}$ has an induced 4-cycle iff $n, m \geq 2$, so Theorem 7 is the case $X = \mathbf{n} \multimap \mathbf{m}$ and $Y = \mathbf{n}' \multimap \mathbf{m}'$. The curried type $(\mathbf{2} \multimap \mathbf{2}) \multimap ((\mathbf{2} \multimap \mathbf{2}) \multimap \mathbf{2})$, which Theorem 7 does not cover, is an instance[L49].

7 Local Certainty and Duality

The deterministic hull mixes states that are certain on all events at once. A weaker requirement is certainty within each of a family of contexts. This section places the coherence space between the deterministic hull and the resulting local relaxation. It ends with a quantum event logic, which has no certain states.

7.1 The Local Relaxation

Definition 7 (local relaxation [3][L50]). *For a given finite set E of events and a family $\mathcal{C}$ of subsets of E, the* contexts*, the* local relaxation *is the set of $x : E \to \mathbb{R}_{\geq 0}$ such that $\sum_{e \in C} x_e \leq 1$ for every context C. The* deterministic hull *is* Mix *of the elements of the local relaxation with values in $\{0, 1\}$. The* normalized *local relaxation and deterministic hull are the sets defined in the same way with $=$ in place of $\leq$.*

The local relaxation contains its $\{0, 1\}$-valued elements and is closed under mixtures, so Definition 3 applies to it, with the deterministic hull as Mix of its certain states. The normalized local relaxation and deterministic hull are the general probabilistic models and the classical models of a family of contexts [3].

For a formula A, take as contexts the supports of the certain states of $A^\perp$. The pairing with a $\{0, 1\}$-valued function is the sum over its support, so the local relaxation of A is the orthogonal of the certain states of $A^{\perp}$[L51]. We write $\mathrm{loc}(A)$ for it, and we prove that it bounds the coherence space from above.

Theorem 9 (local certainty[L51,L34,L16]). *For every formula A, the deterministic hull of A is contained in $P(A)$, which is contained in* $\mathrm{loc}(A)$*. Moreover, $P(A) = \mathrm{loc}(A)$ iff $A^\perp$ has no gap. At T the second inclusion is an equality, and if X, Y, X', Y' have at least two elements each, the first is strict.*

An orthogonal does not change when a set is replaced by its mixtures, so $\mathrm{loc}(A)$ is the orthogonal of the deterministic hull of $A^\perp$. By Definition 5(iii), $P(A)$ is the orthogonal of $P(A^\perp)$. The deterministic hull of $A^\perp$ is contained in $P(A^\perp)$, which gives $P(A) \subseteq \mathrm{loc}(A)$. The deterministic hull and the coherence space of every formula are equal to their biorthogonals, the deterministic hull by the bipolar theorem in the proof of Theorem 5, which gives the equivalence. At T the dual is the core, which has no gap (Theorem 5(i)), and the first inclusion is strict by Theorem 3.

7.2 A Clique Outside the Coherence Space

The converse of the last statement of Theorem 1 fails: some cliques of the coherence space are not certain states. Let $X = Y = X' = Y' = \mathrm{Bool}$. On the five points of Theorem 3, two distinct points cohere in T iff they are not exclusive. Order them as $c_0, \ldots, c_4$ along the cycle of length 5 of the exclusivity graph. The coherence graph on them is the complement of this cycle, again a cycle of length 5. Two distinct points of $T \otimes T$ are incoherent iff they are incoherent in one component, so a clique of $(T \otimes T)^\perp$ contained in $\{c_0, \ldots, c_4\}^2$ is a stable set of the strong product of two cycles of length 5. The set $\{(c_i, c_{2i}) \mid i \in \mathbb{Z}/5\}$ is such a stable set, of size 5, Shannon's stable set [41]. We show that it gives a clique outside the coherence space.

Theorem 10 (a clique outside the PCS[L52,L53,L54]). *Shannon's set is a clique of the coherence space $(T \otimes T)^\perp$, and its indicator does not lie in $P((T \otimes T)^\perp)$.*

The function z of Theorem 3, with value $\frac{1}{2}$ on the five points, lies in $P(T)$, so $z \otimes z$ lies in $P(T \otimes T)$. Its pairing with the indicator of Shannon's set is $5 \cdot \frac{1}{4} = \frac{5}{4} > 1$.

The indicator of every clique of A lies in $\mathrm{loc}(A)$. The certain states of $A^\perp$ are cliques of $A^\perp$ (Theorem 1). A clique of A and a clique of $A^\perp$ share at most one point, since two distinct points cannot cohere in both. This allows us to prove that Shannon's set also separates the coherence space from the local relaxation.

Corollary 2 (the second inclusion[L51]). *At $(T \otimes T)^\perp$, the coherence space is strictly contained in the local relaxation.*

We did not find the placement of the coherence space between the deterministic hull and the local relaxation (Theorem 9) in the literature.

7.3 Negation, Tightness, and the Tensor

Two sets U and V of weight vectors on one web form a *tight pair* if each is the orthogonal of the other, $U = V^\perp$ and $V = U^\perp$ [30]. A coherence space and its orthogonal form a tight pair. The question is when the deterministic hulls of a formula and of its negation form a tight pair, and we prove the following answer.

Theorem 11 (duality[L55,L56]). *Let A be a formula. (i) The orthogonal of the deterministic hull of A is $\mathrm{loc}(A^\perp)$, and the orthogonal of $\mathrm{loc}(A)$ is the deterministic hull of $A^\perp$. (ii) The deterministic hulls of A and of $A^\perp$ form a tight pair iff the deterministic hull of A is $\mathrm{loc}(A)$, iff neither A nor $A^\perp$ has a gap. (iii) If X, Y, X', Y' have at least two elements each, the deterministic hulls of the core K and of T do not form a tight pair.*

Part (i) combines the biorthogonal closure of the deterministic hull with the fact that an orthogonal does not change when a set is replaced by its mixtures. Each of the two equations of a tight pair of deterministic hulls implies the other, since the deterministic hull is equal to its biorthogonal. Part (ii) follows with Theorem 9, and (iii) with Theorem 5(i) and the gap at T (Theorem 3).

When the certain states of A and of $A^\perp$ are the cliques of their coherence graphs, (i) is the antiblocking identity between the stable set polytope of a graph and the clique-constrained polytope of its complement, which joins two distinct vertices iff the graph does not [23,34]. In that case tightness holds iff the coherence graph is perfect [40,14]. For a coherence space E, let $p(E)$ be the set of $x \geq 0$ with total weight at most 1 on every clique of $E^\perp$. Then $p(E^\perp) = p(E)^\perp$ when E is generated from $\mathbf{1}$ by $\perp$ and &, so that its coherence graph is a cograph, and not when its coherence graph has an induced cycle of length 5 [19]. A family of finite sets and its orthogonal family have dual norm balls iff they are the cliques and the stable sets of a perfect graph [9]. These results concern the cliques of a coherence space. Theorem 11 concerns its certain states, which can be fewer than the cliques (Theorem 10). To our knowledge, part (ii), the equivalence of tightness with the absence of a gap at A and at $A^\perp$, does not appear in the literature.

We show that tightness holds at first order and fails at a tensor of first-order types.

Corollary 3 (tightness and the tensor[L57]). *For all n and m, the local relaxation of $\mathbf{n} \multimap \mathbf{m}$ is $P(\mathbf{n} \multimap \mathbf{m})$, so the deterministic hulls of $\mathbf{n} \multimap \mathbf{m}$ and of its negation form a tight pair. For $n, m, n', m' \geq 2$, the deterministic hulls of $(\mathbf{n} \multimap \mathbf{m}) \otimes (\mathbf{n}' \multimap \mathbf{m}')$ and of its negation do not.*

The indicator of a row $\{a\} \times \{1, \ldots, m\}$ is a certain state of the negation of $\mathbf{n} \multimap \mathbf{m}$. These indicators cut out the row-substochastic matrices, which form $P(\mathbf{n} \multimap \mathbf{m})$. With Proposition 1 the deterministic hull is the local relaxation. The second statement is Theorem 11 with the gap of Theorem 3. So the deterministic hull is not compositional: tightness at A and at B does not imply tightness at $A \otimes B$. The coherence space is compositional by construction.

7.4 Bell's Theorem at the Core

Let $X = Y = X' = Y' = \mathbf{2}$. A point $((x, a), (y, b))$ of the web of the core K is the event that two parties with inputs x and y give outputs a and b. Two events are *locally orthogonal* if some party has equal inputs and different outputs. For distinct events this is the negation of exclusivity (Section 4.2). An element of $P(K)$ is *normalized* if it has total weight 1 for each pair of inputs. We prove the following form of Bell's theorem [6] at the core.

Theorem 12 (Bell at the core[L58]). *Every certain state of T is the indicator of a set of pairwise locally orthogonal events. The Popescu–Rohrlich box [47], with value $\frac{1}{2}$ on the events with $a + b \equiv xy \pmod 2$ and 0 elsewhere, lies in $\mathrm{loc}(K)$ and not in $P(K)$.*

Two events that are not locally orthogonal are both produced by one deterministic product strategy, a pair of deterministic programs, one for each party, which lies in $P(K)$ and has pairing 2 with their indicator. No three events in the support of the box are pairwise locally orthogonal, so the box has total weight at most 1 on the support of every certain state of T, and lies in $\mathrm{loc}(K)$. By Theorem 5, $P(K)$ is the deterministic hull of K. Every certain state of K lies below a deterministic product strategy, which has total weight at most 3 on the eight events with $a + b \equiv xy$, the CHSH inequality [15]. The box has total weight 4 there. By Theorem 9, $P(K) \neq \mathrm{loc}(K)$ iff T has a gap.

The normalized elements of $P(K)$ are the Bell-local behaviours, i.e. the mixtures of deterministic product strategies that always terminate. A set of pairwise locally orthogonal events meets each clique of K in at most one event, so $\mathrm{loc}(K)$ is the set of functions with total weight at most 1 on every such set. Its normalized elements are the no-signalling behaviours [22]. These two identifications are not formalized.

7.5 Quantum Event Logic

In quantum logic, the events are the closed subspaces of a Hilbert space [7]. A context is an orthogonal basis, a maximal set of pairwise orthogonal rays. We take

as events the 18 rays of Cabello's set in $\mathbb{R}^4$, spanned by vectors with coordinates in $\{0, \pm 1\}$, and as contexts its 9 orthogonal bases, each ray lying in exactly two of them [11][L59,L60]. We write $1\bar{1}00$ for the ray spanned by $(1, -1, 0, 0)$. A *state* is an element of the normalized local relaxation of Definition 7. For a unit vector ψ, the *Born rule* gives the ray spanned by v the value $x_v = (\psi \cdot v)^2/(v \cdot v)$. This is a state[L64,L63]. The mixtures of these states are the *quantum states*.

The construction of Section 2 gives the empty deterministic hull here: no state has values in $\{0, 1\}$ [36,11], so the normalized deterministic hull is empty. We formalize this known fact[L61,L62]. We also prove that the normalized local relaxation contains states that are not mixtures of Born-rule assignments of these rays.

Theorem 13 (local certainty in quantum logic[L65]). *On Cabello's set, the function h with value $\frac{1}{2}$ on the nine rays $\bar{1}111$, 0001, 0010, 0100, $1\bar{1}\bar{1}1$, $1\bar{1}1\bar{1}$, $11\bar{1}1$, $111\bar{1}$, 1111 and 0 elsewhere is a state and is not a mixture of Born-rule assignments.*

Every basis contains exactly two of the nine rays, so h is a state. The ray 0001 lies in the plane of the orthogonal rays 1001 and $100\bar{1}$, so every Born-rule assignment satisfies $x_{0001} \leq x_{1001} + x_{100\bar{1}}$, the difference being ψ_1^2. The function h violates this law, with values $\frac{1}{2}$, 0, and 0. No basis contains 0001 and either of the other two rays, while each condition of the local relaxation involves a single basis.

In Theorem 13, each ray v is represented by the projection onto v in $\mathbb{R}^4$. A state x has a *quantum model* if some Hilbert space has a projection P_v for each ray v, summing to the identity on each basis, and a positive operator ρ of trace 1 with $x_v = \mathrm{tr}(\rho P_v)$ for every ray v. Whether every state on Cabello's set has a quantum model was left open [3]. The state h has none in finite dimension. It is the only state that vanishes outside its nine rays, so it is an extreme point of the set of states. This step is not formalized. An extreme state with a value outside $\{0, 1\}$ has no quantum model [50]. On all rays of $\mathbb{R}^d$ with $d \geq 3$, every state is a quantum state [26].

8 Related Work

Probabilistic semantics. In quasi-Borel spaces, a model of higher-order probability, every distribution is the image of a distribution on the real line under a map into the space [29]. The elements of a probabilistic coherence space are not given in this form.

Generalized probabilism. The construction of Section 2 is generalized probabilism [43,53]. The laws valid on all truth assignments to a set of events are Boole's conditions of possible experience [45]. Other derivations obtain the sum and product rules of probability from properties of a lattice of events [17,52,35]. In these approaches the events are formulas or lattice elements. For a formula, the events are the points of its web, which carry no connectives.

Probabilistic theories and contextuality. Generalized probabilistic theories describe a system by its states and its effects, the outcomes of measurements [4,46], and under the no-restriction hypothesis every map from states to

$[0, 1]$ that respects mixtures is an effect [31]. Contextuality has combinatorial formulations by exclusivity graphs [12], by hypergraphs of contexts [3], and by sheaves [2]. Its noncontextual models form correlation polytopes [44], and it also occurs in other domains [1]. In these formulations the scenario is given by its measurements and outcomes. Here, we work with webs of a type, with exclusivity defined by coherence.

Causal processes. Probabilistic coherence spaces model the deep-inference logic BV, which adds a noncommutative connective for sequential composition [8]. Higher-order causal theories, built from categories of processes compatible with a causal order [32], also model BV. Over matrices of nonnegative numbers they give this model of probabilistic coherence spaces restricted to normalized maps [49].

9 Conclusion

At first order the coherence space of a type is the set of mixtures of deterministic programs. At second order it also contains behaviours that no randomized sequential program has, and induced four-cycles decide when this happens. By duality the gap is equivalent to Bell's theorem at the dual type. The deterministic hull is not compositional. The coherence space is, at the cost of these behaviours. So a law of sequential programs, such as the KCBS inequality, need not hold in the model.

These results bear on three uses of the model. First, arguments that replace a randomized program by a distribution over deterministic ones, as in Yao's minimax principle [55], are valid in the model exactly at the types without a gap, such as the first-order types, and not at T over Bool. Second, a bound proved over a coherence space holds for every program of its type, and the deterministic hull shows when it is tight. On the five points of the pentagon, the coherence space allows total weight $\frac{5}{2}$, while every sequential program has at most 2. Third, if the environments of a categorical framework for composable security [10] are interpreted in coherence spaces, they include non-sequential ones, so security proved against them is stronger than security against programs.

Open questions include other models of probabilistic programs, an exact condition beyond second order, where the coherence graphs are not cographs, and the exponential types. The theta body of a graph lies between its stable set and clique-constrained polytopes and gives the quantum bounds of a contextuality scenario [12], so it may describe programs with quantum resources. A comparison with categories of processes compatible with a causal order [32] is also open.

Data Availability Statement

The Lean 4 development that mechanizes every result marked with a superscript number such as $^{\text{L1}}$ is provided as an anonymized archive with this submission. It builds with `mathlib`, contains no `sorry`, and uses only the axioms `propext`, `Classical.choice`, and `Quot.sound`. It will be made publicly available and submitted for artifact evaluation.

Disclosure of Generative AI Use

A generative artificial intelligence (AI) assistant (Claude, Anthropic) was used to draft and edit the text, to search the literature, and to develop parts of the Lean formalization. The authors checked every statement, proof, citation and Lean result, and take full responsibility for the content.

References

1. Abramsky, S.: Relational databases and Bell's theorem. In: In Search of Elegance in the Theory and Practice of Computation, LNCS, vol. 8000, pp. 13–35. Springer (2013)
2. Abramsky, S., Brandenburger, A.: The sheaf-theoretic structure of non-locality and contextuality. New Journal of Physics **13**, 113036 (2011). https://doi.org/10.1088/1367-2630/13/11/113036
3. Acín, A., Fritz, T., Leverrier, A., Sainz, A.B.: A combinatorial approach to nonlocality and contextuality. Communications in Mathematical Physics **334**(2), 533–628 (2015). https://doi.org/10.1007/s00220-014-2260-1
4. Barrett, J.: Information processing in generalized probabilistic theories. Physical Review A **75**(3), 032304 (2007). https://doi.org/10.1103/PhysRevA.75.032304
5. Baumeler, Ä., Wolf, S.: The space of logically consistent classical processes without causal order. New Journal of Physics **18**, 013036 (2016). https://doi.org/10.1088/1367-2630/18/1/013036
6. Bell, J.S.: On the Einstein Podolsky Rosen paradox. Physics Physique Fizika **1**(3), 195–200 (1964)
7. Birkhoff, G., von Neumann, J.: The logic of quantum mechanics. Annals of Mathematics **37**(4), 823–843 (1936)
8. Blute, R., Panangaden, P., Slavnov, S.: Deep inference and probabilistic coherence spaces. Applied Categorical Structures **20**, 209–228 (2012)
9. Borodulin-Nadzieja, P., Farkas, B., Pelczar-Barwacz, A.: Geometric duality, perfect graphs, and the Sierpiński space (2026), arXiv:2605.14072
10. Broadbent, A., Karvonen, M.: Categorical composable cryptography. In: Foundations of Software Science and Computation Structures (FoSSaCS 2022). LNCS, vol. 13242, pp. 161–183. Springer (2022)
11. Cabello, A., Estebaranz, J.M., García-Alcaine, G.: Bell–Kochen–Specker theorem: A proof with 18 vectors. Physics Letters A **212**(4), 183–187 (1996)
12. Cabello, A., Severini, S., Winter, A.: Graph-theoretic approach to quantum correlations. Physical Review Letters **112**(4), 040401 (2014)
13. Castellan, S., Clairambault, P., Paquet, H., Winskel, G.: The concurrent game semantics of probabilistic PCF. In: Proceedings of the 33rd Annual ACM/IEEE Symposium on Logic in Computer Science (LICS). pp. 215–224 (2018)
14. Chvátal, V.: On certain polytopes associated with graphs. Journal of Combinatorial Theory, Series B **18**(2), 138–154 (1975)
15. Clauser, J.F., Horne, M.A., Shimony, A., Holt, R.A.: Proposed experiment to test local hidden-variable theories. Physical Review Letters **23**(15), 880–884 (1969)
16. Corneil, D.G., Lerchs, H., Stewart Burlingham, L.: Complement reducible graphs. Discrete Applied Mathematics **3**(3), 163–174 (1981)
17. Cox, R.T.: Probability, frequency and reasonable expectation. American Journal of Physics **14**(1), 1–13 (1946)

18. Danos, V., Ehrhard, T.: Probabilistic coherence spaces as a model of higher-order probabilistic computation. Information and Computation **209**(6), 966–991 (2011)
19. Ehrhard, T.: MPRI 2–2, TD 2. Exercise sheet, IRIF, https://www.irif.fr/~ehrhard/pub/td2021-2.pdf (2021)
20. Ehrhard, T., Pagani, M., Tasson, C.: Full abstraction for probabilistic PCF. Journal of the ACM **65**(4), 23:1–23:44 (2018)
21. de Finetti, B.: Theory of Probability, vol. 1. Wiley (1974)
22. Fritz, T., Sainz, A.B., Augusiak, R., Brask, J.B., Chaves, R., Leverrier, A., Acín, A.: Local orthogonality as a multipartite principle for quantum correlations. Nature Communications **4**, 2263 (2013). https://doi.org/10.1038/ncomms3263
23. Fulkerson, D.R.: Blocking and anti-blocking pairs of polyhedra. Mathematical Programming **1**, 168–194 (1971)
24. Girard, J.Y.: Linear logic. Theoretical Computer Science **50**(1), 1–101 (1987)
25. Girard, J.Y.: Between logic and quantic: A tract. In: Ehrhard, T., Girard, J.Y., Ruet, P., Scott, P. (eds.) Linear Logic in Computer Science, London Mathematical Society Lecture Note Series, vol. 316, pp. 346–381. Cambridge University Press (2004)
26. Gleason, A.M.: Measures on the closed subspaces of a Hilbert space. Journal of Mathematics and Mechanics **6**(4), 885–893 (1957)
27. Golumbic, M.C.: Trivially perfect graphs. Discrete Mathematics **24**, 105–107 (1978)
28. Grötschel, M., Lovász, L., Schrijver, A.: Geometric Algorithms and Combinatorial Optimization. Springer (1988)
29. Heunen, C., Kammar, O., Staton, S., Yang, H.: A convenient category for higher-order probability theory. In: 32nd Annual ACM/IEEE Symposium on Logic in Computer Science (LICS). pp. 1–12. IEEE (2017)
30. Hyland, M., Schalk, A.: Glueing and orthogonality for models of linear logic. Theoretical Computer Science **294**(1–2), 183–231 (2003). https://doi.org/10.1016/S0304-3975(01)00241-9
31. Janotta, P., Lal, R.: Generalized probabilistic theories without the no-restriction hypothesis. Physical Review A **87**(5), 052131 (2013). https://doi.org/10.1103/PhysRevA.87.052131
32. Kissinger, A., Uijlen, S.: A categorical semantics for causal structure. Logical Methods in Computer Science **15**(3), 15:1–15:48 (2019). https://doi.org/10.23638/LMCS-15(3:15)2019
33. Klyachko, A.A., Can, M.A., Binicioğlu, S., Shumovsky, A.S.: Simple test for hidden variables in spin-1 systems. Physical Review Letters **101**(2), 020403 (2008)
34. Knuth, D.E.: The sandwich theorem. Electronic Journal of Combinatorics **1**, A1 (1994). https://doi.org/10.37236/1193
35. Knuth, K.H., Skilling, J.: Foundations of inference. Axioms **1**(1), 38–73 (2012). https://doi.org/10.3390/axioms1010038
36. Kochen, S., Specker, E.P.: The problem of hidden variables in quantum mechanics. Journal of Mathematics and Mechanics **17**(1), 59–87 (1967)
37. Kozen, D.: Semantics of probabilistic programs. Journal of Computer and System Sciences **22**(3), 328–350 (1981)
38. Kuhn, H.W.: Extensive games and the problem of information. In: Contributions to the Theory of Games II, Annals of Mathematics Studies, vol. 28, pp. 193–216. Princeton University Press (1953)
39. Leblanc, H., Morgan, C.G.: Probabilistic semantics for intuitionistic logic. Notre Dame Journal of Formal Logic **24**(2), 161–180 (1983)
40. Lovász, L.: Normal hypergraphs and the perfect graph conjecture. Discrete Mathematics **2**(3), 253–267 (1972)

41. Lovász, L.: On the Shannon capacity of a graph. IEEE Transactions on Information Theory **25**(1), 1–7 (1979)
42. de Moura, L., Ullrich, S.: The Lean 4 theorem prover and programming language. In: Automated Deduction – CADE 28. LNCS, vol. 12699, pp. 625–635. Springer (2021)
43. Paris, J.B.: A note on the Dutch book method. In: Proceedings of the Second International Symposium on Imprecise Probabilities and Their Applications (ISIPTA). pp. 301–306 (2001)
44. Pitowsky, I.: Correlation polytopes: Their geometry and complexity. Mathematical Programming **50**, 395–414 (1991). https://doi.org/10.1007/BF01594946
45. Pitowsky, I.: George Boole's 'conditions of possible experience' and the quantum puzzle. The British Journal for the Philosophy of Science **45**(1), 95–125 (1994). https://doi.org/10.1093/bjps/45.1.95
46. Plávala, M.: General probabilistic theories: An introduction. Physics Reports **1033**, 1–64 (2023). https://doi.org/10.1016/j.physrep.2023.09.001
47. Popescu, S., Rohrlich, D.: Quantum nonlocality as an axiom. Foundations of Physics **24**(3), 379–385 (1994)
48. Ravindra, G.: Perfectness of normal products of graphs. Discrete Mathematics **24**(3), 291–298 (1978). https://doi.org/10.1016/0012-365X(78)90100-0
49. Simmons, W., Kissinger, A.: Higher-order causal theories are models of BV-logic. In: 47th International Symposium on Mathematical Foundations of Computer Science (MFCS). LIPIcs, vol. 241, pp. 80:1–80:14 (2022)
50. Sujan V.K., Kunjwal, R.: No-go theorem for quantum realization of extremal correlations (2025), arXiv:2509.14879
51. The mathlib Community: The Lean mathematical library. In: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020). pp. 367–381. ACM, New York, NY, USA (2020)
52. Van Horn, K.S.: Constructing a logic of plausible inference: A guide to Cox's theorem. International Journal of Approximate Reasoning **34**(1), 3–24 (2003)
53. Williams, J.R.G.: Generalized probabilism: Dutch books and accuracy domination. Journal of Philosophical Logic **41**(5), 811–840 (2012)
54. Wolk, E.S.: The comparability graph of a tree. Proceedings of the American Mathematical Society **13**, 789–795 (1962)
55. Yao, A.C.C.: Probabilistic computations: Toward a unified measure of complexity. In: 18th Annual Symposium on Foundations of Computer Science (FOCS). pp. 222–227. IEEE (1977)